

Summary Sheet

Pruned evidence-gated pharmacogenomic LLM-alert capstone

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Conference / Symposium

Field	Value
Name	Machine Learning for Health Symposium (ML4H 2026)
URL	https://ml4h.ahli.cc/
Author instructions / CFP	https://ml4h.ahli.cc/submit/call-for-papers/
Submission deadline	September 10, 2026, 11:59 PM AoE
Author notification	October 22, 2026
Camera-ready deadline	November 7, 2026 (tentative)
Event dates	December 6-7, 2026
Location	Sydney, Australia
Suggested track	Findings Track for preliminary specification-conformance work; Proceedings Track only after independent review.
Editable paper format	LaTeX/Overleaf source: paper/ml4h_findings_evidence_gate.tex with paper/ml4h_findings_refs.bib; compiled PDF: paper/ml4h_findings_evidence_gate.pdf.

Title and Abstract

Title: Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims: Detecting Overclaiming Relative to Guideline-Supported Evidence

Abstract: Large language models can draft medication alerts whose fluency exceeds the evidence they cite. This capstone studies one workflow: pharmacogenomic and genomic medication-alert text. I implement a synthetic pre-display evidence gate with three checks: citation/guideline support, population fit, and endpoint/actionability. Stage 1 does not call an LLM or parse free text; it tests a deterministic controller over structured annotations. In one-gate ablations, disabling citation/guideline support, population fit, or endpoint/actionability allowed 2/23, 1/23, and 6/23 designed overclaims through unchanged, respectively. With all gates active, all 23 designed overclaim archetypes were blocked or narrowed and all 10 bounded alerts passed unchanged. Primary-gate conformance was 31/33, with two gate-precedence disagreements reported as negative results. These are specification-conformance results, not clinical safety, generalization, or patient-benefit evidence.

Generative AI Use

Generative AI tools were used for brainstorming, critique, code scaffolding, drafting, packaging, and formatting. The author selected the final scope, directed the pruning, verified sources, reviewed generated text, ran local tests, and remains responsible for all claims, limitations, and submission materials.

Optional Data, Code, and Tools

- Synthetic 33-case pharmacogenomic/genomic medication-alert case bank.
- Executable three-gate controller and tests.
- One-gate ablation CSV showing isolated necessity of citation/guideline support, population fit, and endpoint/actionability gates.
- CSV/JSON results for overclaim detection, inappropriate denial, calibration, and gate/error categories.

- No real patient data, no protected health information, no diagnosis, and no treatment recommendation.